MILITARY SPECIFICATION

GENERATOR SETS, DIESEL ENGINE DRIVEN, 15 THRU 200 KILOWATTS 50/60 AND 400 HERTZ, (TACTICAL)

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for tactical generator sets (hereinafter referred to as "set" or "sets") from 15 kilowatts (kW) thru 200 kW, 50/60 Hertz (Hz) and 400 Hz, alternating current (ac), skid-mounted, diesel engine driven, liquid-cooled.

1.2 Classification. The sets shall be Type I (Tactical) and of the following Classes, Modes, and Sizes as specified on the applicable detail specification (see 3.1 and 6.2):

Class 1 - Precise.
Class 2 - Utility.
Mode I - 50/60 Hz.
Mode II - 400 Hz (Class 1 only).
Size 15 - 15 kW at 60 Hz (12.5 kW at 50 Hz).
Size 30 - 30 kW at 60 Hz (25 kW at 50 Hz).
Size 60 - 60 kW at 60 Hz (50 kW at 50 Hz).
Size 100 - 100 kW at 60 Hz (83.3 kW at 50 Hz).
Size 200 - 200 kW at 60 Hz (167 kW at 50 Hz).

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1.3 Set rating. The generator sets shall be rated 0.8 power factor (pf), lagging, and shall be reconnectable for the following voltage ratings:

(a) 3 phase, 4-wire, 240 volts line-to-neutral and 416 volts line-to-line (240/416 volts).
(b) 3 phase, 4 wire, 120 volts line-to-neutral and 208 volts line-to-line (120/208 volts).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft Missile and Ordnance.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-A-8421	- Air Transportability Requirements, General Specification for.
MIL-E-10062	- Engine, Preparation for Shipment and Storage of.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservation and Break-in.
MIL-G-28554	- Generator Sets, Mobile Electric Power; Packaging of.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

(See supplement 1 for list of associated specification sheets (detail specification).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-471	- Maintainability Demonstration.
MIL-STD-705	- Generator Sets, Engine-Driven, Methods of Tests and Instructions.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-831	- Test Reports, Preparation of.
MIL-STD-1400	- Military Standard Engines, Gasoline and Diesel, Methods of Test.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits for Army Materiel.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2101 L Street, NW, Washington DC 20037.)

3. REQUIREMENTS

3.1 Detail requirements for individual set size, class, and mode. The individual set requirements shall be as specified herein and in accordance with the applicable detail specifications. In the event of any conflict between the requirements of this specification and the detail specifications, the latter shall govern.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed drawing requirements is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide

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tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the set. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.1.2 Reliability. Reliability shall be as indicated in the detailed specification.

3.2 First article (preproduction model). The contractor shall furnish one generator with excitation system and two complete assembled sets for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that he is capable of producing sets that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.5). When specified (see 4.6.1, 4.7.1, and 6.2), the Government may conduct any of the preproduction examination and tests.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more sets for inspection as specified in 4.4.

3.4 Starting, operating, handling, and storage.

3.4.1 Starting. The set shall start (see 6.4.1) as follows.

3.4.1.1 Without winterization. The set shall start within 5 minutes:

(a) At ambient temperatures from +125° F (51.7° C) to -25° F (-31.7° C) at any possible relative humidity.
(b) With 12 inches of rain per hour impinging on the set at angles from the vertical up to 45 degrees.
(c) With the base of the set in planes from level to 15 degrees from level.
(d) At any altitude up to and including 5,000 feet above sea level at an ambient temperature of 107° F (41.7° C).
(e) At 8,000 feet above sea level at an ambient temperature of 95° F (35° C).

3.4.1.2 With integral fuel burning winterization system. At ambient temperatures from -25° F (-31.7° C) to -65° F (-53.9° C), the set shall start:

(a) Within 60 minutes, with use of the integral fuel burning winterization system for a maximum of 55 minutes and external batteries (see 6.4.16) not warmer than -25° F (-31.7° C) used only for cranking.

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(b) Within 5 minutes, after a 24-hour period of standby operation (see 6.4.3) using the integral fuel burning winterization system and with external batteries (see 6.4.16) not warmer than -25° F (-31.7° C) used only for cranking.

3.4.1.3 With integral electric winterization system. At ambient temperatures from +125° F (51.7° C) to -65° F (-53.9° C) and after 24 hours of standby operation (see 6.4.3) using the integral electric winterization system, the set shall start and accept 75 percent of rated load within 20 seconds without the use of any starting aids except for external batteries (see 6.4.16) used only for cranking.

3.4.2 Operating. Immediately after starting, the set shall run at no-load for 15 minutes (maximum warm-up period) except no warm-up period is permitted for the requirements of 3.4.1.3. At the end of this warm-up period, the set shall operate as specified herein except that failures in the reliability test during the warm-up period are applicable to reliability calculations (see 3.1). Servicing, adjustments, and replacements of parts shall be limited to the scheduled maintenance specified in table I, and the unscheduled maintenance as permitted by the reliability and maintainability requirements as specified in the detail specification (see 3.1).

3.4.2.1 Diesel fuel. Using the specified diesel fuel (see 3.5.3.1), the set shall start and operate, at continuous loads up to and including rated load, under the conditions specified in 3.4.1.1 for a total of 5,000 hours, except that at the 8,000 foot altitude condition (see 3.4.1.1(e)), rated load shall be 90 percent of the set rating, unless otherwise specified in the detail specification (see 3.1).

3.4.2.2 Turbine fuel. Using the specified JP-4 fuel (see 3.5.3.2), the set shall be capable of starting and operating, at continuous loads up to and including 85 percent of the set rating for a minimum period of 200 hours, at ambient temperatures from 0° F (-17.8° C) to 100° F (37.8° C) (see 3.4.1.1).

3.4.2.3 Operating speed. Operating (rated) speed for the sets shall be as specified in the applicable detail specification (see 3.1). Sets shall operate at 2500 rpm for a period of 5 minutes without damage (see 6.4.17).

3.4.3 Stopping. Within 30 seconds after activation of any device intended to stop the set, the set shall stop (see 6.4.2).

3.4.4 Lifting and towing. The two lifting attachments shall carry a minimum combined weight of eight times the total weight of the set with the vertical distance from the top of the set to the point of lift equal to the set width. The set shall remain within 15 degrees of level when lifted by a crane. The towing provisions shall withstand a minimum of five times the total weight of the set.

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TABLE I. Servicing, adjustments, and replacement schedule.

Item No.	Minimum Hours Of Operation	Servicing, Adjustment, and Replacement Allowed
1	8	Check lubricating oil level, add if necessary. If running, perform without stopping engine.
2	8	Check coolant level, add if necessary. If running, perform without stopping engine.
3	8	Check batteries.
4	100	Inspect engine for loose connections; leaks in oil, fuel, and water system; and free action of all moving parts. Service as necessary and record.
5	100	Clean air filter element.
6	300	Change lubricating oil and filters.
7	100	Change fuel oil filters.
8	100	Clean crankcase breather.
9	100	Clean electric fuel pump strainers.
10	300	Replace air filter element.
11	300	Adjust belt tension.
12	100	(Class 1 sets only) Check hydraulic fluid level and add hydraulic fluid as necessary.
13	500	Check valve tappet clearance and reset if necessary.
14	500	Take compression pressures at cranking speeds.
15	500	Clean injector nozzles, adjust fuel setting.
16	500	(Class 1 sets only). Change hydraulic fluid and hydraulic filter. Clean governor actuator filter.
17	500	Adjust governor and throttle linkage, excluding actuator on class 1 sets.
18	500	Clean and grease battery terminals and cable lugs.
19	1000	Replace fuel injectors.
20	1000	Clean radiator exterior cooling surfaces.
21	1000	Replace belts.
22	1000	Adjust governor actuator on class 1 sets.

3.4.5 Rough handling. The set shall not be damaged (see 6.4.9) by rough handling which could be encountered during rail, truck, aircraft, and helicopter transportation (see 6.4.10, 6.4.11, and 6.4.12).

3.4.6 Storage temperature and humidity. The set shall not be damaged (see 6.4.4) by storage in temperatures from +155° F (+68.3° C) to -65° F (-53.9° C) at any relative humidity possible for temperature within that range and shall not be damaged by exposure to five consecutive 48-hour cycles as specified in MIL-STD-705, Method 711.1.

3.4.7 Audio noise. The set shall not exceed the audio noise limits specified in the detailed specification (see 3.1).

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3.4.8 Air transportability characteristics. The set shall meet the air transportability requirements of MIL-A-8421 when the set is loaded into the aircraft with the longitudinal axis of the set (radiator facing either front or rear) parallel to the longitudinal axis of the aircraft.

3.4.9 Flexural vibrations and critical speeds. The set shall be free from dangerous flexural vibrations (see 6.4.5) and danagerous torsional critical speeds (see 6.4.8) between 1200 and 2500 rpm.

3.5 Lubricating oils, fuels, and coolant. Lubrication, fuel, and coolant shall be as specified herein.

3.5.1 Lubricating oil. The engine lubricating oil shall conform to MIL-L-2104 as defined by the appropriate lubrication order, when ambient temperatures are above -10° F (-23° C). The engine lubricating oil for use in the range of 0 to -65° F (-18° to -54° C) shall conform to MIL-L-46167.

3.5.2 Hydraulic oil. For class 1 sets, the hydraulic oil used with the governor system shall conform to MIL-H-5606.

3.5.3 Fuels.

3.5.3.1 Diesel fuel. Diesel fuel when required (see 3.4.2.1), shall conform to VV-F-800, military symbol DF-A, DF-1, or DF-2, as applicable.

3.5.3.2 Turbine fuel. JP-4 fuel, when required (see 3.4.2.2), shall conform to MIL-T-5624.

3.5.4 Coolant. Cooling compound shall conform to MIL-A-11755.

3.6 Engine. Except as otherwise specified herein, the engine, including all systems, components, accessories, and auxiliaries, shall conform to applicable drawings and MIL-STD-1410, class IV.

3.6.1 Horsepower. The minimum continuous brake horsepower shall equal or exceed the value specified in the applicable detail specification (see 3.1) when operating on diesel fuel (see 3.5.3.1).

3.6.2 Cranking system. The cranking system shall be 24 volts dc negative ground. The set shall not be damaged (see 6.4.17) in the event polarity of the battery cables is reversed. Batteries with electrolyte shall be furnished in the preproduction sets. Dry charged batteries without electrolyte shall be furnished as part of the individual production sets.

3.6.3 Engine safety devices. The engine safety devices as specified herein and in the applicable detail specification, shall act to simultaneously stop the set, open the interrupter and to cause the appropriate indicating lamp of the malfunction indicator to energize. The overspeed safety device shall trip between 2400 and 2500 rpm of the engine. The high coolant temperature safety device shall trip at 222° F ± 3° F. The low fuel level safety device shall trip when the fuel supply in the day tank has dropped to a point at which it contains sufficient fuel to operate the set at rated load for a minimum of 1 minute.

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3.6.4 Louver control. When the temperature of the engine liquid coolant is 160° F (71.1° C) and below, the vanes of the exhaust air louver assembly shall be closed.

3.6.5 Air cleaner condition device. This device shall energize the indicating light on the control panel when the pressure drop across the air cleaner element exceeds 20 ± 1.5 inches of water.

3.6.6 Starting aid. Ether tanks shall be furnished in the preproduction sets. Individual production sets shall be furnished less ether tanks.

3.7 Governing system. The governing system shall provide the frequency and speed performance, as specified in the applicable detail specification (see 3.1).

3.8 Generator and excitation system. The generator and excitation system (exciter, voltage regulator, and other accessories necessary to control the output voltage) shall meet the requirements herein and of the applicable detail specification (see 3.1).

3.8.1 Generator. The generator shall withstand operation at 2500 rpm for 15 minutes without damage (see 6.4.17).

3.8.2 Temperature rise. The temperature rise shall not exceed 90 C° (162 F°) for class B insulation and 100 C° (180 F°) for class F insulation, when measured by the resistance method.

3.8.3 Insulation resistance. Insulation resistance shall be not less than 1 megohm in an ambient temperature of 77° F (25° C).

3.8.4 Dielectric strength. Windings shall withstand the following 60 Hz voltages applied for 1 minute:

(a) Generator armature - 1,832 volts between phase windings and 1,480 volts between phase windings and ground.
(b) Generator field and exciter windings - 10 times ceiling voltage, but neither less than 1,500 nor more than 3,500 volts, (applied between windings and ground).
(c) Windings energized by the 24 volt dc control, cranking and battery charging systems - 500 volts, (applied between windings and ground).
(d) All others - twice rated voltage plus 1000 volts, (applied between winding and ground and between windings where applicable).

3.8.5 Short circuit. The generator and excitation system shall withstand 10 second single phase line-to-line, single phase line-to-neutral and symmetrical 3 phase short circuits applied at its output terminals, when operating at rated load, without reduction of the dielectric strength to a point where it will not meet the requirements of 3.8.4. The sustained short circuit current shall be not less than 300 percent of rated output current.

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3.8.6 Winding resistance. The resistance of each winding shall be as specified on the applicable drawing. All resistance values shall be corrected to a temperature of 77° F (25° C).

3.8.7 Voltage unbalance. Maximum difference between line-to-line voltages shall be not more than 5 percent of rated voltage for the 240/416 volt connection under the condition of a single-phase, line-to-line, unity-pf load of 25 percent of rated current and no other load on the generator.

3.8.8 Phase balance (voltage). The maximum difference in the three line-to-neutral voltages shall be not more than 1 percent of rated line-to-neutral voltage. The maximum difference between the voltages of the 120-volt windings of any one phase shall be not more than 1 volt.

3.8.9 Voltage waveform. The deviation factor for the line-to-line and line-to-neutral voltage waveform for each voltage connection shall be not more than 5 percent, nor shall any single voltage harmonic be more than 2 percent. In addition, there shall be no evident discontinuities, spikes, or notches in the waveform when viewed on an oscilloscope having a bandwidth of DC - 1.5 MHz and a useable viewing screen of 8 x 10 cm. The oscilloscope gain shall be adjusted such that one cycle of voltage wave covers approximately the entire viewing screen.

3.8.10 Saturation curves. The curves listed herein shall be plotted with the generator armature terminal volts on the vertical axis versus the exciter field current on the horizontal axis:

(a) Open circuit saturation curve.
(b) Synchronous impedance curve (short-circuit saturation curve).
(c) Zero power factor saturation curve.
(d) Rated load current saturation curve.

The exciter field current shall be as specified in the detailed specification (see 3.1).

3.8.11 Voltage modulation. The voltage modulation for each of the line-to-line, line-to-neutral, and Coil T9-T12 voltages for any load from no-load to rated load shall not exceed 1 percent.

3.8.12 Phase sequence (rotation). Phase rotation shall be L_1, L_2, and L_3.

3.8.13 Voltage reconnection system. Generator sets shall be shipped connected for 120/208 volt operation.

3.9 Parallel operation. The requirements for parallel operation are as specified in the applicable detail specification (see 3.1).

3.10 Instruments, controls, and other devices. Instruments, controls, and devices shall meet the requirements herein (also see 3.1) when mounted in the set and with the set operating.

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3.10.1 Protective devices. Protective devices shall act to open the main contactor, and cause the appropriate malfunction indicator lamp (see 3.10.2) to light. When specified herein, the device shall also act to stop the engine.

3.10.1.1 Short circuit. This device shall operate instantaneously in the event the set output is short circuited with either a single phase or a 3 phase fault.

3.10.1.2 Overvoltage. This device shall act to stop the engine. It shall actuate in not more than 1-1/4 seconds after the voltage across Coil T9-T12 has risen to and remained at any value greater than 153 ± 3 volts for not less than 200 milliseconds.

3.10.1.3 Overload. This device shall not trip when current in all three phases is less than 110 percent of rated value; it shall trip within 8 ± 2 minutes for 130 percent of rated current in any phase.

3.10.1.4 Reverse power. This device shall actuate if power flow into the set exceeds 20 ± 3 percent of rated value.

3.10.2 Malfunction indicator. The malfunction indicator system shall cause the appropriate indicating lamp to energize, under action of the protective devices (see 3.6.3 and 3.10.1).

3.10.3 Instrument accuracy. The maximum allowable error/accuracy of the metering system shall be as follows. (For definition of accuracy/error see 6.4.13.)

Instrument	Maximum Allowable Error/Accurracy
Voltmeter:	
115-125 volt range	5 volts
200-250 volt range	10 volts
Ammeter (ac)	6 percent
Kilowattmeter	10 percent
Frequency meter	1.0 percent

3.11 Treatment and painting. The set shall be treated and painted in accordance with the applicable drawings (see 3.1), except as specified herein. The following shall not be painted: Terminal wiring connections, governor linkage, instruction diagrams, nameplates, rectifiers, relays, circuit breakers, switches, hose, drivebelts, and all other parts whose operation or function would be adversely affected by paint. Treatment and painting shall be optional for the following if located within the set housing and not visible during operation of the set: Corrosion-resisting metals; metals treated to be corrosion resistant; and items treated and painted to be corrosion resistant by the original manufacturers, such as MS starting motor, ether tank, main contactor. After assembly and testing, the interior of the set shall be

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retouched to correct any paint damage and the entire exterior of the set, except items adversely affected, shall be oversprayed.

3.12 Information and instruction plates. Preproduction sets shall be delivered with all specified plates installed. For production sets, information and instruction plates may be installed after individual quality conformance tests (see 4.5) have been completed and the final overspray of paint is applied (see 3.11).

3.13 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the set shall not exceed the UMO4 limits of class C2 equipment of MIL-STD-461 even though the set kVA rating may be greater than that specified in the standard.

3.14 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of any inspection by the preproduction generator, generator with excitation system, or sets shall be cause for disassembly, in the presence of a Government representative, to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction set. Reassembly with acceptable components or parts and reinspection shall be the responsibility of the contractor.

4.1.2 Inspection failure. Should the Government elect to perform or repeat any inspection or test in this specification, failure of a set to meet any requirement specified herein shall be cause for refusal by the Government to accept production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the Government at no cost to the Government on any sets previously produced under the contract. Any deficiencies found as the result of such elective inspection or test will be considered prima facie evidence that all sets accepted prior to the completion of such inspection or test are similarly deficient unless evidence to the contrary is furnished by

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the contractor and such evidence is acceptable to the contracting officer. The provisions of this paragraph apply, notwithstanding any prior acceptance of first articles or first article test reports.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article (preproduction) inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.8).
(e) Inspection of packaging (see 4.9).

4.3 First article (preproduction) inspection. The preproduction generator (with excitation system) and sets shall be examined and tested as specified herein to determine compliance with this specification and the applicable detail specification (see 3.1). Noncompliance shall be cause for performing the disassembly inspection specified in 4.1.1.

4.3.1 Examination of components and subassemblies. Examination of components and subassemblies shall be made prior to assembly of the generator or set. Evidence that any components or subassemblies do not comply with the requirements of the drawings shall be cause for rejection of that component or subassembly.

4.3.1.1 Examination of generator, excitation system and sets. Examination of the generator with excitation system and the two sets shall be made without disassembly. Evidence that the generator, excitation system, or sets do not comply with the drawings shall be cause for rejection of the generator, excitation system, or sets.

4.3.2 Tests. Conduct the applicable tests marked "X" in column 2 of table II. The set tests may be conducted in any order, except that the first two tests shall be "Railroad impact" and "Humidity" in that order, and the last test shall be "Endurance-reliability." Failure of any test to comply with the applicable requirement paragraph cited in column 7 of table II shall be cause for performing the inspection specified in 4.1.1.

4.3.3 Inspection report. The contractor shall furnish, within the time interval specified (see 6.2), an inspection report in accordance with MIL-STD-831 covering preproduction inspection. The inspection report shall also include the following:

(a) All test data obtained from applicable tests.
(b) Description of malfunction, damages, failures, and adjustments (other than adjustments permitted by this specification) which occur during inspection.
(c) Causes and analysis of malfunction, damages, or failures, and reasons for adjustments.
(d) Corrective actions taken or required.

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4.4 Initial production inspection. When specified (see 3.3), one or more initial production sets will be selected at random by the Government from the sets being produced with production tooling and will be subjected to the tests marked "X" in column 3 of table II to determine conformance to the requirements of this specification. These sets shall have successfully passed the examination and tests specified in 4.5.1. The inspection will be performed by the Government at a site selected by the Government, utilizing the appropriate technical manual for maintenance and servicing procedures. Acceptance of an initial production set shall not exclude the remaining sets from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production tests, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production set(s) and shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all sets accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Individual set inspection.

4.5.1.1 Examination. Each set shall be examined, without disassembly for conformance to the drawings (see 3.1). Noncompliance with the drawings shall be cause for rejection.

4.5.1.2 Tests. Each set shall be subjected to the applicable tests marked "X" under column 1 of table II. For generator only tests, the tests shall be conducted prior to assembly into the set. Failure of any test shall be cause for rejection.

4.5.2 Sample inspection. Sampling shall be in accordance with MIL-STD-105 after all sets comprising a lot have passed the inspection specified in 4.5.1. Unless otherwise specified (see 6.2), lot sizes shall consist of not more than 50 sets except that the first lot shall contain not less than 25 sets. The inspection level shall be S-3 and the AQL shall be 6.5 percent defective. No adjustment or substitution of components shall be made on the sets selected for sample inspection.

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4.5.2.1 Tests. Sets selected in accordance with 4.5.2 shall be subjected to the tests marked "X" under column 4 of table II in the order listed with the exception that the endurance test shall be conducted on 50 percent of the samples from each lot. When this 50 percent results in fractions of sets, the next highest number representing complete sets shall be used. A sample set which fails to meet a requirement during any test shall be considered a defective set.

4.6 Test schedules. The test schedules shall be as shown herein. Nonconformance to the applicable requirement paragraph shown in column 7 of the schedules shall constitute failure of the test and rejection of the generator, the generator with excitation system, or the set, as applicable. The Government reserves the right to reject the equipment for not meeting any requirement herein, even though not performing a test directly related to the specific requirement. The requirements in Section 3 shall apply to all tests performed as a part of another test. If maintenance shutdown, as permitted in table I, servicing, adjustments, and replacement schedule, would be due to occur (because of hours accrued during previous testing) after the beginning of any test, the shutdown shall be postponed until completion of the test or until completion of 100 hours of operation, whichever is the lesser.

4.6.1 Table II. Unless otherwise specified (see 6.2), the test schedule shall be as shown in table II (see 4.7.1).

4.6.2 Table III, schedule of optional tests. The optional tests as indicated in table III shall be conducted by the contractor to determine set compliance subsequent to changes in parts or material when changes are determined major changes by the contracting officer. All quality assurance provisions of Section 4 shall apply.

TABLE II. Test schedule.

Indiv.	First Art. (Pre-prod.)	IPT	Sample	Test	Test Method (MIL-STD-705), Test Paragraph, or As Required Below	Requirement Paragraph
1	2	3	4	5	6	7
X	X	-	X	Engine	4.7.11	3.6
				Generator Only		
X	X	-	-	Winding resistance	401.1	3.8.6
X	X	-	-	High potential	302.1 (isolate diodes)	3.8.4
X	X	-	-	Insulation resistance	301.1 (isolate diodes)	3.8.3

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TABLE II. Test schedule. (Cont'd)

1	2	3	4	5	6	7
				Generator with Excitation System		
-	X	-	-	Overspeed (generator only)	505.3	3.8.1
-	X	-	-	Open-circuit saturation curve	410.1 (At all rated frequencies)	3.8.10
-	X	-	-	Synchronous impedance curve (Short-circuit saturation curve)	411.1 (At all rated frequencies)	3.8.10
-	X	-	-	Zero power factor saturation curve	412.1 (At all rated frequencies)	3.8.10
-	X	-	-	Rated load current saturation curve	413.1 (At all rated frequencies)	3.8.10
-	X	-	-	Direct-axis transient reactance	425.1	3.1
-	X	-	-	Negative sequence reactance and impedance	422.1	3.1
-	X	-	-	Short circuit (mechanical strength)	625.1 (Apply single phase line-to-line, single phase line-to-neutral and symmetrical 3 phase short circuits each for 10 seconds at the output terminals.)	3.8.5
				Generator Set		
-	X	-	-	Railroad impact	740.5, 4.7.4	3.4.5
-	X	-	-	Humidity	711.1 (See note 11)	3.4.1.1
X	X	-	-	Start and stop	503.1	3.4.1, 3.4.3
X	X	-	-	Frequency and voltage regulation, stability and transient response (short term)	608.1a (At all rated frequencies) (See notes 11 and 12)	3.7, 3.8

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TABLE II. Test schedule. (Cont'd)

1	2	3	4	5	6	7
-	X	X	-	Frequency and voltage stability (long term)	608.2 (At all rated frequencies)	3.7, 3.8
X	X	-	-	Overspeed (set)	505.1 (At 2500 rpm for 5 minutes)	3.4.2.3
X	X	-	-	Overspeed protective device	505.2	3.6.3
X	X	-	X	Phase balance (voltage)	508.1 (To be measured at load terminals. Also, measure voltages for each winding of each phase using separate excitation. On individual production sets, measure line-to-line and line-to-neutral at load terminals only.)	3.8.8
X	X	-	-	Circuit interrupter (short circuit)	512.1 (Use oscillograph for preproduction model only, all single phase L_1-L_0, L_2-L_0, L_3-L_0. For production sets, L_1-L_0 only.)	3.10.1.1
X	X	-	-	Circuit interrupter (overload current)	512.2	3.10.1.3
X	X	-	-	Circuit interrupter (overvoltage and undervoltage)	512.3 - (Class 1 sets - both both over and under voltage. Class 2 sets - over voltage only)	3.10.1.2
X	X	-	-	Indicating instrument (electrical)	513.1 (For all voltage connections)	3.10.3
X	X	-	-	Underspeed protective device	506.1a (At all rated frequencies)	3.10

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TABLE II. Test schedule. (Cont'd)

1	2	3	4	5	6	7
-	X	-	X	Reverse battery polarity	516.5	3.6.2
X	X	-	-	Low oil pressure protective device	515.1	3.6.3
X	X	-	-	Over temperature protective device	515.2	3.6.3
X	X	-	-	Low fuel protective device	515.5	3.6.3
X	-	-	-	Regulator range	511.1 (At all rated frequencies)	3.8
X	X	-	-	Phase sequence (rotation)	507.1	3.8.12
X	X	-	-	Frequency adjustment range	511.2 (At no-load and rated load. At all rated frequencies)	3.7
-	X	X	-	Voltage unbalance with unbalanced load	620.2	3.8.7
-	X	-	-	Lifting and towing	740.4	3.4.4
-	X	-	X	Voltage waveform (Oscillographic) (Harmonic Analysis)	601.1 and 601.4 (At all rated voltage and frequency connections and the following loading conditions) (See Note 3) (a) No-load (b) Rated load at 0.8 pf lagging (c) Rated load at 1.0 pf	3.8.9
-	X	-	X	Voltage modulation	602.1 (Perform test at no-load and rated load, all line-to-line and line-to-neutral portions and coil T9 to T12.)	3.8.11
-	X	-	X	Voltage dip for low power factor load	619.1 (See Note 3)	3.8
-	X	-	-	Voltage dip and rise for rated load	619.2	3.8

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TABLE II. Test schedule. (Cont'd)

1	2	3	4	5	6	7
X	X	-	-	Parallel operation	630.1 and 4.7.3 (First article sets to be tested at all rated frequencies.) (For Production sets see note 2.)	3.9, 3.10.1.4
-	X	-	X	High temperature at 125° F (51.7° C)	710.1 (Take temperature rise measurements on stator windings and exciter field under the following conditions: (See Note 4) MODE I sets 50 Hz, 380 volts 50 Hz, 426 volts 60 Hz, 395 volts 60 Hz, 480 volts MODE II sets 400 Hz, 395 volts 400 Hz, 458 volts) (Also perform 619.1 rated frequency and voltage).(See Note 11)	3.8.2
X	X	-	-	Maximum Power (For gasoline and diesel generator sets)	640.1 (Preproduction sets) and 640.4 (Production sets)	3.6.1
X	X	-	-	DC control	655.1 (Perform at minimum and maximum settings of alternator voltage adjustment control) (See Note 14 and 15.)	3.6.2
-	X	X	X	Endurance-reliability	4.7.10 (See Note 5, 6, 8, 9, 10, 11 and 13)	3.1, 3.4.2

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TABLE III. Schedule of optional tests.

Test	Test Method (MIL-STD-705), Test Paragraph, or As Required Below	Requirement Paragraph
1	2	3
	Generator Set	
Rain	711.3 (To include 45 degrees)	3.4.1.1
Inclined operation	660.1	3.4.1.1
Fuel consumption	670.1	3.1
Starting and operating (Moderate cold-Battery Start)	701.2 at -25° F (-31.7° C) (See Note 11)	3.4.1.1
Motor starting	4.7.9	3.8, 3.1
Audio noise	4.7.2	3.4.7
Altitude operation (Before Endurance - reliability test)	720.1 (At 8000 feet, 95° F (35° C) and 5000 feet, 107° F (41.7° C)) (See Note 11)	3.4.1.1
Endurance-reliability	4.7.10 (See Note 7, 9, 10 and 11)	3.1
Maintainability demonstration	4.7.8, 4.7.1	3.1
Altitude operation (After Endurance - reliability test)	720.1 (At 8000 feet, 95° F (35° C) and 5000 feet, 107° F (41.7° C)) (See Note 11)	3.4.1.1
Road	(For 15 through 100 kW Sets only Trailer mount set and subject to conditions in 6.4.11)	3.4.5
Drop (ENDS)	740.3, 4.7.5	3.4.5
Air transportability	4.7.7	3.4.5, 3.4.8
Electromagnetic Interference	4.7.6 (At all rated voltage and frequency connections at the following load conditions: (a) no-load (b) rated kW load at 1.0 PF (c) rated kW load at 0.8 PF	3.13
Starting and operating (Extreme cold battery start)	701.1 at -65° F (-53.9° C)	3.4.1.2
Torsiographing	504.2 (At NO LOAD and rated load at all rated frequencies)	3.4.9
High temperature at 125° F (51.7° C)	710.1 (Take temperature rise measurements on rotor windings only) (slip rings req'd)	3.8.2

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Special Notes:

1. Test shall be conducted with the 240/416 volt connection, unless otherwise noted.
2. Test on mode I generator and set shall be conducted at 60 Hz only, unless otherwise indicated.
3. Also view waveform with an oscilloscope (see 3.8.9). For sample tests, only method 601.4 need be used.
4. For sample tests, only make the runs which gave hightest stator and exciter field temperature rises during the preproduction test.
5. For sample sets, endurance runs shall be 200 hours. The disassembly inspection contained in method 690.1 is not required.
6. The duration of the Endurance-reliability test shall be a minimum of 1000 hours for each preproduction and initial production set.
7. The duration of the endurance-reliability tests shall be a minimum of 2500 hours. The set shall run 200 hours on JP-4 fuel after the first 200 hours have been run on VV-F-800 fuel.
8. One of the preproduction sets shall run 200 hours on JP-4 fuel after the first 200 hours have been run on VV-F-800 fuel.
9. For mode I, class 2 sets during the Endurance-reliability test, Method 690.1, the pre-endurance 608.1 test, paragraph 690.1.3.2.1(a), shall be conducted with the frequency regulation adjusted between 2.7 and 3 percent of rated frequency. For the endurance test, paragraph 690.1.3.2.2, the frequency regulation may be adjusted between 1 and 3 percent of rated frequency, providing the set meets the stability requirements of the detailed specification (see 3.1). The adjustment may be made at (1) the start of the endurance test, (2) prior to operation on JP-4 fuel, (3) at the conclusion of the JP-4 operation, and (4) at each 500 hour interval. When conducting the 608.1 of paragraph 690.1.3.2.2(c), if the frequency transient performance exceeds the specified requirements of the detailed specification, then record the final value of frequency regulation and repeat the 608.1 (rated load) test with the frequency regulation adjusted between 2.7 and 3 percent. If compliant, readjust to the previously recorded value of frequency regulation and perform the scheduled maintenance.
10. During the Endurance-reliability test, conduct test at 480 v for a minimum of 200 hours for the 60,100 and 200 kW sets.
11. For 200 kW sets only, whenever Method 608.1 is specified, in lieu of the test load conditions (608.1.3.2(c)) of 4/4, 3/4, 2/4, 1/4 and 4/4, substitute the following:
    (a) Three cycles (load application and removal) between each pair of test load conditions listed below: No load (NL) and 1/2 rated load, NL and 1/4 rated load, and 1/4 and 3/4 rated load.
    (b) Three cycles of load change between NL and rated load to verify that the circuit interrupter does not trip.
    (c) Where method 608.1 at rated load only is specified, substitute 1/4 to 3/4 rated load.
12. For definition of rated load see 1.2, 1.3, 3.4.2.1, 3.4.2.2 and 6.4.7.

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13. The set fuel tank shall be used as the fuel source for conducting the sample endurance test. The tank shall be inspected for leaks during test and after completion of test. The tank shall be completely drained after the final inspection.
14. Three volts is required between maximum and minimum voltage. Set control at midrange after test.
15. For individual tests, only verify that sets will operate without batteries and note results on data sheet. Instrumentation not required for individual tests.

4.7 Test procedures. Tests shall be conducted as specified herein. Test instruments shall be of the laboratory type and shall have been calibrated within 30 days of the start of testing and at 6-month periods thereafter. Instruments used in calibration should have at least five times the accuracy of the instrument being calibrated. Calibrated reference instruments of lesser accuracy than standard and which are not used for any other purpose may be used for the required periodic check of test instruments. Instruments shall be calibrated at the frequencies at which they are going to be used. Direct-reading instruments shall have at least 0.5 percent instrument manufacturer's accuracy and shall be connected to indicate in the most accurate portion of their range. On dc instruments, the readings shall not be made on the lower 15 percent of the scale. On ac instruments, the readings shall not be made on the lower one-third of the scale. When the test methods call for the use of voltage and frequency recording-type meters, other than oscillographs, the Texas Instrument Company Model PDRHXFHXVA-A16-XT shall be used. Oscillograph galvanometer frequency response shall be flat (plus or minus 5 percent) from DC to not less than 3,000 Hz. When recording meters are specified for any part of a test, turn on the recording meters prior to starting the warmup period of the set and record continously for the duration of the test. Unless otherwise specified, the recording meters shall be operated at a minimum speed of 6 inches per hour during the portions of the test where steady-state loading conditions exist and shall be operated at a minimum speed of 6 inches per minute at least 30 seconds before, during, and after a load change. Individual tests requiring a series of data points shall be repeated if an interruption occurs during the test (e.g.; data points being taken for plotting a curve, etc.).

4.7.1 Government performed tests. The maintainability demonstration and road test will be accomplished by the Government at its expense at a Government installation.

4.7.2 Audio noise test. Except as specified herein, the instrumentation and procedures for the audio noise test shall conform to MIL-STD-1474. When any component is changed which may affect the noise limits (see 3.1) the number of feet to be specified in the caution notice as required in MIL-STD-1474 shall be determined by selecting the measurement position with the highest sound pressure levels and repositioning the microphone outward to a point where the measured sound pressure levels fall within the specified limits. Measure the distance from the set to the new microphone position and repeat the test for the eight positions at the new distance. All tests shall be conducted at NO-LOAD and RATED LOAD.

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4.7.3 Parallel operation test for individual sets. Each production set shall be tested in parallel with one of the first production sets (see 6.2) randomly selected by the Government, in accordance with method 630.1 with the following changes:

(a) Control panel instrumentation of the sets shall be used.
(b) Load conditions shall be no-load, half load and rated load.
(c) Where 1 hour of operation is referenced, substitute 15 minutes of operation.

4.7.4 Railroad impact test. The railroad impact test shall be performed in accordance with method 740.5 with the following changes:

(a) Paragraph 740.5.3 (a), (c), (e) and 740.4 (b), delete "generator" and substitute "generator sets."
(b) Paragraphs 740.5.3(b) and (f), delete "Method 614.1, Voltage and Frequency Regulation Test" and substitute, "Method 608.1, Frequency and Voltage Regulation, Stability and Transient Response Test, on each set, at rated load only."

4.7.5 Drop test. The drop test shall be performed in accordance with Method 740.3 with the following changes:

(a) Paragraphs 740.3(a) and (g), delete "Method 614.1, Voltage and Frequency Regulation Test" and substitute, "Method 608.1, Frequency and Voltage Relulation, Stability and Transient Response Test, on each set, at rated load only."
(b) Paragraph 740.3.5, add "(e) Voltage and frequency performance requirements."

4.7.6 Electromagnetic interference test. The electromagnetic interference test shall be performed in accordance with MIL-STD-462.

4.7.7 Air transportability test. The set shall be tested in accordance with MIL-A-8421 to determine compliance with the air transportability requirements.

4.7.8 Maintainability demonstration test. The maintainability demonstration shall be performed in accordance with method 1B of MIL-STD-471 with assigned risk of 10 percent (ϕ = 1.28).

4.7.9 Motor starting test. The motor starting test shall be performed by use of a motor rated NEMA Code F, in accordance with MG-1. The motor shall be loaded with a flywheel having an inertia equal to that of the motor rotor. Satisfactory starting is defined as acceleration of the motor to rated speed without tripping any safety devices.

4.7.10 Endurance-reliability test. The endurance-reliability test shall be conducted in accordance with test method 690.1 of MIL-STD-705 and MIL-STD-781, Test Plan IV.

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4.7.11 Engine tests. The engines shall be tested in accordance with MIL-STD-1410, class IV. In lieu of the production control test specified in 5.3 of MIL-STD-1410, perform the production control test specified in MIL-STD-1400, test method series 3000. Nonconformance to 5.2 of MIL-STD-1410 or test method series 3000 of MIL-STD-1400 shall constitute failure.

4.8 Inspection comparison. The Government may select sets at any time during the contract production period and subject these sets to any examination and test specified herein, necessary to determine that the selected sets meet all requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Sets will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction models and initial production sets. Acceptance of an inspection comparison set shall not exclude the remaining sets from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.8.1 Inspection failure. Failure of an inspection comparison set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.8 shall be cause for rejection of the inspection comparison set and shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all sets accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.9 Inspection of packaging. The preservation, packing and marking shall be examined and tested in accordance with the Quality Assurance Provisions of MIL-G-28554, unless otherwise specified in the detail specification (see 3.1).

5. PACKAGING

5.1 Preservation, packing and marking. Each complete generator set shall be preserved, packed, and marked in accordance with MIL-G-28554, unless otherwise specified in the detail specification (see 3.1). The level of preservation shall be A, B, or C and the level of packing shall be A, B, or C as specified (see 6.2). Generator sets preserved level C shall have the engine fuel systems, combustion chambers, and valves preserved in accordance with MIL-E-10062, type II, level A, method I. The crankcase shall be filled to operating level with preservative lubricating oil conforming to MIL-L-21260, type I, grade 30, except for 200 kW generator sets, type II, grade 30, shall be used.

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5.1.1 Replacement of preservative lubrication oil. The preservative lubrication oil shall be drained and replaced when engines have completed 100 hours of operation. A tag shall be prepared for each crankcase indicating: "This crankcase is filled to the operating level with preservative lubricating oil good for 100 hours of operation - Do not drain until running time meter indicates 100 hours - Check Oil Level - If low, elevate the operating level with the operating oil (MIL-L-2104, Applicable Grade)." The tag shall be attached to the crankcase fill tube.

5.1.2 Preservation of hydraulic sump. The hydraulic sump for the Governing system shall be filled to the operating level with hydraulic oil conforming to MIL-H-5606 for class I sets.

5.2 Connection tag. A connection tag shall be attached to one of the load terminals, stating: "This set is connected for 120/208 volts".

6. NOTES

6.1 Intended use. The sets are intended to supply power for multipurpose use in military application. The general characteristics of the sets are described in MIL-STD-633 and MIL-STD-1332.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Detail specification title, number, and date (see 3.1).
(c) Time frame required for submission of first article preproduction model (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of sets to be furnished, when applicable (see 3.3).
(f) Time interval for submission of first article preproduction test reports (see 4.3.3).
(g) Size of lots for production sets, if other than as specified (see 4.5.2).
(h) When test schedule is other than as specified in Table II (see 4.6.1).
(i) Level of preservation and level of packing required (see 5.1).
(j) How the paralleling sets are to be retained and maintained during life of contract and how the sets are to be refurbished and delivered at the end of the contract (see 4.7.3).

6.3 Optional equipment. Optional items may be ordered separately but will not be installed on generator set (see MIL-STD-633).

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6.4 Definitions.

6.4.1 Start. A set is considered to have started when it is operating at rated voltage and speed without the further use of starting aids.

6.4.2 Stop. A set is considered to have stopped when all rotating members are at zero rpm.

6.4.3 Standby operation. A condition during which a nonoperating set is kept warm and ready to start through use of winterization systems.

6.4.4 Temperature and humidity damage. Temperature and humidity damage is defined as conditions causing malfunction of any component or part, corrosion, breakage, deformation, or reduction of insulation resistance below 50,000 ohms.

6.4.5 Dangerous flexural vibration. Dangerous flexural vibration is defined as a vibration which occurs at a speed at which maximum stress in the shaft from flexural vibration exceeds 9,000 psi.

6.4.6 Stable engine operating conditions (frequency). Stable engine operating conditions are the conditions specified for short term steady-state performance.

6.4.7 Rated load. Rated load is rated kW at rated power factor, rated frequency (speed) and rated voltage.

6.4.8 Dangerous torsional critical speed. Dangerous torsional critical speed is defined as the speed at which maximum vibrating stress in the shaft from torsional vibration exceeds 5,000 psi.

6.4.9 Rough handling damage. Rough handling damage is defined as any condition resulting in malfunctioning of the set, deformation, loosening, breakage, or change of fit of any component or part.

6.4.10 Normal railroad transportation. Normal railroad transportation shall be interpreted to mean impact speeds up to and including 10 miles per hour.

6.4.11 Normal truck or trailer transportation. Normal truck or trailer transportation is defined as the conditions encountered during four cycles of a road test, each cycle consisting of the following, with the set mounted on a suitable Army truck or trailer:

Road Condition	Distance (miles)	Speed (mph)
Paved highway	250	up to 50
Level cross country	250	up to 20
Hilly cross country	125	up to 20
Belgian block	15	up to 20

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6.4.12 Normal aircraft and helicopter transportation. Normal aircraft and helicopter transportation shall be interpreted to mean a 12-inch end drop.

6.4.13 Accuracy. Accuracy is a ratio which defines the limit of error expressed as a percentage of full-scale value. Error is the difference between the indication and the true value of the quantity measured. It is the quantity which, when algebraically subtracted from the indication, gives the true value. A positive error denotes that the indication of the meter is greater than the true value.

6.4.14 Failure. A failure is defined as the inability of the set(s) to meet the requirements herein. A failure classification board shall be established by the contracting officer to identify and classify all failures.

6.4.14.1 Relevant failure. A relevant failure is any failure of an item which prevents the set from meeting the output requirements specified herein. Relevant failures shall be used to compute set reliability.

6.4.15 Inspection. Inspection is the examination and testing of supplies or services including, when appropriate, raw materials, components, and intermediate assemblies to determine conformance with contract requirements.

6.4.16 External batteries. For purposes of demonstrating the requirements specified in this specification, external batteries are defined as two (2) each MS35000-3 or -4 batteries connected in series. They shall be connected to the set by means of a 15 foot slave cable. The slave and battery cable assemblies shall be fabricated with 1/0 wire conforming to MIL-C-5756. Battery terminal adapters shall conform to MS75004. Terminal lug MS25036-133 shall be used to secure the 1/0 wire to the battery terminal adapters. The positive cables shall have a red sleeve for easy identification. One end of the slave cable shall have a connector conforming to MS75057-1.

6.4.17 Damage. Damge is defined as any failure (see 6.4.14), rough handling damage (see 6.4.9) or degradation in life. Fusing of a replaceable fuse is not considered damage, provided it is performing its intended function.

6.5 Preproduction model. Any changes or deviations of production sets from the approved preproduction models or drawings during the production will be subject to the approval of the contracting officer. Approval of the preproduction models will not relieve the contractor of his obligation to furnish sets conforming to this specification.

6.6 Configuration management. The generator sets controlled by this specification are under the Configuration Management authority of the DOD Project Manager for Mobile Electric Power. All Engineering Change Proposals (ECP), Request for Waivers (RFW), and Requests for Deviations (RFD) must be processed in accordance with DLAR 4120.7, AR 700-101, AFR 400-50, NAVMATINST 4120.100A, MC011310.8C, Joint Operating Procedures, Management and Standardization of Mobile Electric Power Generating Sources.

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6.7 Recycled material. It is encouraged that recycled material be used when practical provided the material meets the requirements of the specification and all associated drawings.

6.8 Supersession data. This specification with specification sheets supersedes corresponding military specification (see aplicable detail specification).

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Navy - MC
DLA - GS

Preparing activity:
Army - ME

Project 6115-0226

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